

EPSON OPOS ADK MANUAL

**APPLICATION DEVELOPMENT
GUIDE
MSR**

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Section 1. Introduction

This manual describes the method of use and related items, including device-specific precautions, when the MSR devices are used with EPSON OPOS ADK.

Before the MSR can be used, the EPSON OPOS ADK should be installed and the devices to be used should be set using the SetupPOS utility. For setting methods, please see the Section 2 of this manual.

This manual applies to the following devices.

• EPSON DM-MS123	Can be attached to the SR-610 as options.
• EPSON DM-MX123	Can be attached to the IM-700 as options.
• EPSON DM-MX112	
• EPSON DM-M820	Can be attached to the IM-800 as options.

The operation of MSR is not supported in Windows Vista environment.

Compatibility mode

The compatibility mode for upward compatibility was added in OPOS Ver2.60.

For the details of the compatibility mode, please refer to “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE Compatibility Mode”.

Section 2. Details on Settings

This section describes connection configurations and how to make the settings for MSR.

2.1 Device Information

The DeviceDescription and DeviceName for each model are as follows.

Model Name	DeviceDescription	DeviceName
EPSON DM-MS123	EPSON DM-MS123 MSR	DM-MS123
EPSON DM-MX123	EPSON DM-MX123 MSR	DM-MX123
EPSON DM-MX112	EPSON DM-MX112 MSR	DM-MX112
EPSON DM-M820	EPSON DM-M820 MSR	DM-M820

2.2 Hardware Settings

It is a prerequisite for this OCX to work correctly that headers and footers are set correctly in each device. Before use, use the utilities provided with each device to confirm that the following settings have been made correctly.

2.2.1 EPSON DM-MS123/ DM-M820

Track	Header information	Footer information
1	Yes ("%")	Yes (" ")
2	Yes (",")	Yes (" ")
3	Yes ("+")	Yes (" ")

Read Track ^{*1} : Track1, Track2, Track3

MSR code Type ^{*1} : 101keyboard

^{*1} Before using device, please update the keyboard firmware. Regarding how to update, please refer to the product manual.

2.2.2 EPSON DM-MX123

Track	Header information	Footer information
1	Yes ("%")	Yes ("?"
2	Yes (",")	Yes ("?"
3	Yes ("+"	Yes ("?"

Read Track ^{*1} : Track1, Track2, Track3

MSR code Type ^{*1} : 101keyboard

^{*1} Before using device, please update the keyboard firmware. Regarding how to update, please refer to the product manual.

2.2.3 EPSON DM-MX112

Track	Header information	Footer information
1	Yes ("%")	Yes ("?"
2	Yes (",")	Yes ("?"
3	Yes ("␣")	Yes ("␣R")

Read Track ^{*1} : Track1, Track2, JIS-II

MSR code Type ^{*1} : 101keyboard

^{*1} Before using device, please update the keyboard firmware. Regarding how to update, please refer to the product manual.

2.3 Setting of Devices

After confirming the model number of the hardware, select the correct device using the SetupPOS utility. Regarding how to use the SetupPOS utility, please refer to "EPSON OPOS ADK MANUAL User's Manual (Installer/ SetupPOS/ TMUSB)".

2.4 EPSONHKO (Hook Control Utility)

The MSR device data are all sent to the PC as keyboard data. The hook driver utility is a DLL that separates the data to every device, the MSR, the Keylock, and the Keyboard. When the MSR device's OCX is active (from Open to Close), the hook driver is automatically called up for operation. The incorporation of the EPSONHKO (keyboard hook control utility) also allows the hook control utility to be loaded and operated when the hook driver is required at any other time. The EPSONHKO can be incorporated by the communications setting of the SetupPOS utility. For the EPSONHKO, refer to "EPSON OPOS ADK MANUAL User's Manual (Installer/ SetupPOS/ TMUSB)".

^{*}The hook driver Version 1.xx and Version 2.xx cannot operate simultaneously.

Section 3. Function Details

This section describes the functions of the MSR device in details. Supplementary explanation of the parts not described in detail in "UPOS" is also given here.

3.1 CheckHealth Method

3.1.1 Internal Test

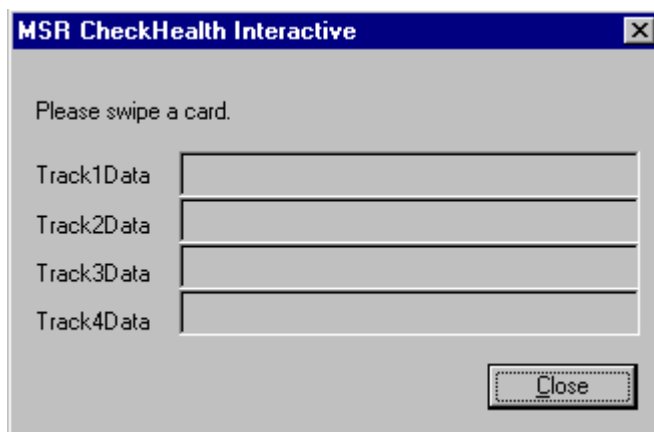
Not supported

3.1.2 External Test

Not supported

3.1.3 Interactive Test

The following dialog box is displayed. When this dialog is displayed, CheckHealth waits for the card to be read. If the card is read at this point, the contents of the Track1Data, Track2Data, Track3Data, and Track4Data properties are displayed in the respective fields. To terminate the test, click the [Close] button.



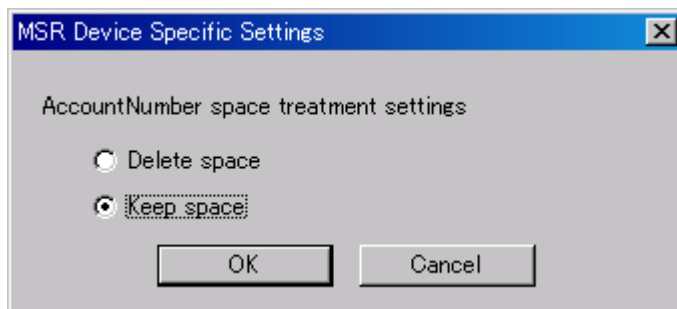
When executed by OPOS_CH_INTERACTIVE, the character strings of the CheckHealthText properties are as follows.

- “Interactive Hcheck: Canceled” : When the [CLOSE] button is pressed while nothing is being displayed in the edit box.
- “Interactive Hcheck: Complete” : When the [CLOSE] button is pressed while something is being displayed in the edit box.

3.2 Device Specific Settings

Some cards may contain space in their Track1Data Property's AccountNumber data.

If the AccountNumber of Track1Data Property should include space, the user may choose to set to eliminate such space in the “Device Specific Settings” dialog box of SetupPOS utility.



[Delete space] : Eliminate space in account number before setting it as AccountNumber Property.

[Keep space] : Do not eliminate space in account number before setting it as AccountNumber Property.

Default setting is [Keep space].

3.3 Property Set Values and Default Values

3.3.1 Capability Set Value

3.3.1.1 EPSON DM-MS123/ DM-M820

CapISO	TRUE
CapJISOne	TRUE
CapJISTwo	FALSE
CapPowerReporting	0 (OPOS_PR_NONE)
CapTransmitSentinels	TRUE
CapWritableTracks	MSR TR NONE

3.3.1.2 EPSON DM-MX123

CapISO	TRUE
CapJISOne	TRUE
CapJISTwo	FALSE
CapPowerReporting	0 (OPOS_PR_NONE)
CapTransmitSentinels	TRUE
CapWritableTracks	MSR TR NONE

3.3.1.3 EPSON DM-MX112

CapISO	TRUE
CapJISOne	TRUE
CapJISTwo	TRUE
CapPowerReporting	0 (OPOS_PR_NONE)
CapTransmitSentinels	TRUE
CapWritableTracks	MSR TR NONE

3.3.2 Property Default Set Values and Setting Ranges

3.3.2.1 All Models

TracksToRead	MSR TR 1 2 3
EncodingMaxLength	0
TracksToWrite	MSR TR NONE

3.4 Device Statistics

The DeviceStatistics function is added in response to the compliance of the "UPOS 1.8".

Please refer to the "EPSON OPOS ADK MANUAL APPLICATION GUIDE Device Statistics" of a separate volume for the details of the Device Statistics.

Section 4. Expanded Functions

This section describes the expanded functions of the MSR device.

4.1 DirectIO Function

The usage of the DirectIO method and DirectIOEvent event is described as follows.

4.1.1 DirectIO Method

No use

4.1.2 DirectIOEvent Event

No use

Section 5. Device Specific Programming

This section describes the functions specific to the MSR device.

5.1 Card Data

5.1.1 EPSON DM-MS123/ DM-M820

It is possible to read the Track1, Track2 and Track3 data from ISO cards or JIS-I cards.

Please note JIS-II card is not supported.

5.1.2 EPSON DM-MX123

It is possible to read the Track1, Track2 and Track3 data from ISO cards or JIS-I cards.

Please note JIS-II card is not supported.

5.1.3 EPSON DM-MX112

It is possible to read the Track1 and Track2 data from ISO cards or JIS-I cards. Track3Data Property will not be set up even if an ISO card or JIS-I card that has Track3 is read.

It is also capable of reading JIS-II card. Any data from JIS-II card shall be set to Track4Data Property.

5.2 Track Discrimination Method

The methods for discriminating between tracks are as follows.

- (1) The interval from the first occurring '%' to '?' is defined as Track1 (in the case of a device compatible with reading of Track1 on ISO or JIS-I cards).
- (2) The interval from the first occurring ';' to '?' is defined as Track2 (in the case of a device compatible with reading of Track2 on ISO or JIS-I cards).
- (3) The interval from the first occurring '+' to '?' is defined as Track3 (in the case of a device compatible with reading of Track3 on ISO or JIS-I cards).
- (4) The interval from the first occurring ' ' to '\R' is defined as Track4 (in the case of a device compatible with reading of JIS-II cards).
- (5) If a character with no meaning is located in front or following the data of any track, this is ignored and analysis is continued.
- (6) If a starting code ('%', ';', '+' or ' ') does not appear at least once in the data, an ErrorEvent is generated as an OPOS_E_EXTENDED and OPOS_EMSR_START argument.
- (7) If a terminating code corresponding to the starting code ('%', ';', '+' or ' ') does not appear, an ErrorEvent is generated as an OPOS_E_EXTENDED and OPOS_EMSR_END argument.
- (8) When the number of characters in a track exceeds the maximum number of characters for the particular track, an ErrorEvent is generated as an OPOS_E_FAILURE argument.
- (9) Cases where ' ' appears first will all be regarded as JIS-II cards, and set as Track4 (in the case of devices compatible with reading of JIS-II cards).

Section 6. Error Information

This section describes the error codes that may result from execution of MSR methods. The common properties and methods are described in "EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE GENERAL DEVELOPMENT". Refer to this guide for more information.

6.1 ResultCode List

6.1.1 When Executing Properties

The ResultCode and ResultCodeExtended when properties are executed are as follows.

Property Name	ResultCode	ResultCodeExtended	Meaning
TracksToRoad	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_ILLEGAL	OPOS_EX_BADPROPVAL	Set value is illegal.
DecodeData	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_ILLEGAL	0	Device decode not required.
ParseDecodeData	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i> .
ErrorReportingType	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_ILLEGAL	OPOS_EX_BADPROPVAL	Set value is illegal.
TransmitSentinels	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_ILLEGAL	0	Refer to <i>UPOS Specifications</i> .
TracksToWrite	OPOS_SUCCESS	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_NOTCLAIMED	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_ILLEGAL	OPOS_EX_INCAPABLE	No function

6.1.2 When Executing Properties

The ResultCode and ResultCodeExtended when Methods are executed are as follows.

Property Name	ResultCode	ResultCodeExtended	Meaning
ClearInputProperties	OPOS_E_SUCCESS	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_NOTCLAIMED	0	Refer to <i>UPOS Specifications</i> .
WriteTracks	OPOS_E_CLOSED	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_DISABLED	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_NOTCLAIMED	0	Refer to <i>UPOS Specifications</i> .
	OPOS_E_ILLEGAL	OPOS_EX_INCAPABLE	No function

6.1.3 When An Error Event Occurs

The ResultCode and ResultCodeExtended codes when an error event occurs are as follows.

When ErrorReportingType = MSR_ERT_CARD

ResultCode	ResultCodeExtended	Meaning
OPOS_E_FAILURE	0	Two or more cards were read when issue of DataEvnet is prohibited.
OPOS_E_ILLEGAL	OPOS_EMSR_ACCOUNTNUMBER	The AccountNumber exceeds 20 characters, or the value is different between Track1 and Track2.
	OPOS_EMSR_EXPIRATIONDATE	ExpirationDate value is different for Track1 and Track2.
	OPOS_EMSR_SECONDFIELD	SecondField exceeds 27 characters.
	OPOS_EMSR_SERVICECODE	The ServiceCode value is different between Track1 and Track2.
	OPOS_EMSR_TITLE	Title field is illegal.
OPOS_E_EXTENDED	OPOS_EMSR_END	The data terminating '?' is missing.

When ErrorReportingType = MSR_ERT_TRACK

ResultCode	ResultCodeExtended	Meaning
OPOS_E_FAILURE	0	Two or more cards were read when issue of DataEvnet is prohibited.
OPOS_E_EXTENDED	OPOS_SUCCESS *	No error
	OPOS_EMSR_END *	Terminating code corresponding to the starting code is missing at the end of the track.
	OPOS_E_FAILURE *	Other error

* ResultCodeExtended (Long data) is divided into bytes, and the error code of each track is stored in the respective bytes.

6.2 Remedial Actions for Principal Errors

ResultCodeExtended	Remedial Action
OPOS_EX_BADINF	Normally, this should not occur. Occurs if the INF file or the Registry information of the device to be used is illegal. Using the INF file provided with EPSON OPOS ADK, perform a new setup using the SetupPOS utility.
OPOS_EMSR_KBHOOKOPEN OPOS_EMSR_KBHOOKSTART OPOS_EMSR_FAILURE	Error related to the hook driver. Normally, this should not occur. Should it occur, then use the SetupPOS utility to confirm that the device is a device for which the hook driver can be used.
OPOS_EX_BADPARAM + 1	The parameter is illegal for the method to be used. Confirm the parameter range for the method to be used.

Section 7. Warnings

- An ErrorEvent is generated if an analysis error occurs in just one of all the tracks.
- When a card is read while the FreezeEvents property is set to TRUE, the read data is stored in the buffer. At this point, up to one data can be stored. Accordingly, an ErrorEvent is generated if two or more data are read. The contents of the ErrorEvent event are OPOS_E_FAILURE.
- If an ErrorEvent event is generated, all the data in the buffer is canceled.
- Do not start reading the next card or changing the Keylock until reading of the card is completed (The LED provided on the MSR lights). If the mentioned actions are performed while data is being processed internally in the MSR, the data will be subjected to inference, and the MSR card contents may not be read correctly.
- The following restrictions exist while the hook driver is activated (When MSR is Open, or the hook control utility is activated).

[Restriction1]

If the card is read while CapsLock is set to ON, the numeric data is recognized as the Shift condition values (!, ", #, \$, %, &, ', etc.), and the card's contents will not be processed correctly.

[Restriction2]

If the card is read while the right Shift key is pressed, the numeric data is recognized as the Shift condition values (!, ", #, \$, %, &, ', etc.), and the card's contents will not be processed correctly. (If the left Shift key is pressed while the card is read, the values will still be recognized correctly.)

[Work around]

If problems occur as result of the restrictions described above, remedy as described below.

1. Set CapsLock to OFF.
 2. Do not assign the right Shift key to programmable keys. If it is unavoidable to do this, use the application to create a restriction that prevents reading of the card while the right Shift key is pressed.
- This MSR always converts data to scan codes. Consequently, the DecodeData property cannot be set to FALSE.
 - If a card is read while a large application (Microsoft Access, etc.) is running, a continuous beep sound may be generated, or the MSR data may become garbled. Perform MSR reading operations while the load on the machine is light.
 - When the EPSONHKO (keyboard hook control utility) is not used, the following phenomena will occur. For the EPSONHKO, please refer to "EPSON OPOS ADK MANUAL User's Manual (Installer/ SetupPOS/ TMUSB)".

When the process which initiated the Hook driver is terminated, the MSR may become inoperable.

For example:

- 1) Opens MSR at process 1.
- 2) Opens MSR at process 2, and makes the MSR ready to use by executing Open, Claim, and DeviceEnabled for MSR.
- 3) Closes MSR at process 1.
- 4) Terminates process 1.
- 5) As the result, MSR cannot be used at process 2.

To avoid this problem, be sure not to terminate the first opened OCX when other MSR is opened.

- This MSR device does not support the power status report function.
- The hook driver Version 1.xx and Version 2.xx cannot operate simultaneously.